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Mobile Survey Design Research
Experiment 18: Response Options

Christopher Antoun
Shelley Feuer
Elizabeth Nichols
Erica Olmsted
Lin Wang

IOE 2015 BCase 01 Project Team
Center for Behavioral Science Methods
Research and Methodology Directorate
U.S. Census Bureau
Washington, D.C. 20233

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Abstract

The present study investigated the effect of response option icon size and button size on respondent performance. The study compared (1) small radio buttons or checkboxes (2 mm in diameter/width), (2) large radio buttons or checkboxes (6 mm in diameter/width), (3) large radio buttons or checkboxes embedded in wide buttons covering the response text, and (4) plain wide buttons covering the response text. The findings show that participants made the least number of errors when tapping the large radio buttons or checkboxes than the other designs; and more participants preferred the large radio buttons or checkboxes than the other designs.

Keywords: Mobile Survey, Response Options, Icons, Button Size

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Author contributions:

Christopher Antoun: study conception and design, analysis and interpretation of results, manuscript preparation

Shelley Feuer, Elizabeth Nichols, Erica Olmsted: analysis and interpretation of results, manuscript preparation

Lin Wang: study supervision, analysis and interpretation of results, manuscript preparation

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Introduction

1. Purpose

To investigate ways to design response options for touch in mobile Web surveys.

2. Specific Research Questions

- Do large icons (radio buttons and checkboxes) in response options improve respondent performance compared to small icons?
- Do wide buttons for response options improve respondent performance compared to radio buttons?

Methods

1. Participants

Sixty-one participants (43 females, 18 males; mean age = 68.9 years ($SD = 4.7$)) participated in the study. Detailed demographics are presented in Table 1. Eighty-six percent of participants reported using a smartphone for more than two years, 7% between one and two years, 6% between 6 months and one year, 1% less than six months and one person had missing data. On a scale of 1-5, where higher scores represent more smartphone familiarity, participants reported an average of 3.9 ($SD = 1.0$). The participants had habitual near vision of 20/50 or better (sufficient for reading newspaper) and had no color vision impairment. All participants were from the metropolitan area of Washington DC and were recruited at community centers.

Performance data from 14 participants were unable to be collected by the experiment software due to technical issues, resulting in a final sample of 47 participants for some indicators. Missing data occurred in each of the experimental conditions and should not affect experimental comparisons.

All participants received a Privacy Act Statement that states the purpose of the study, information to be collected, and the authority under which the information is collected. A written consent to data collection was obtained from all participants. The data collection was approved by the U.S. Office of Management and Budget.

Table 1: Participant Characteristics

Gender (number of participants)	
Male	18
Female	43
Age	
Mean (SD)	68.9 (4.7)
Range	59-75
Education (number of participants)	
Less than high school diploma	1
Completed high school	2
Some college, no degree	13
Associate degree	4

Bachelor degree	19
Post graduate degree	22
Hispanic origin (number of participants)	
Yes	3
No	58
Race (number of participants)	
White	45
Black or African American	6
Asian	8

2. Experimental Paradigm

Participants completed a survey using a smartphone. Upon completion of the survey, the participant completed a paper questionnaire about their experience with the survey.

3. Screen Design

There were four response option designs as depicted in Figure 1: conventional controls (CC), larger controls (LC), larger controls enclosed in wide buttons (LC+WB), wide buttons without any controls (WB). In addition, there were three types of questions: 1. choose-one questions with relatively short response options (20 items); 2. choose-one questions with relatively long response options (3 items); and 3. select-all-that-apply questions (3 items). Each question had seven response options.

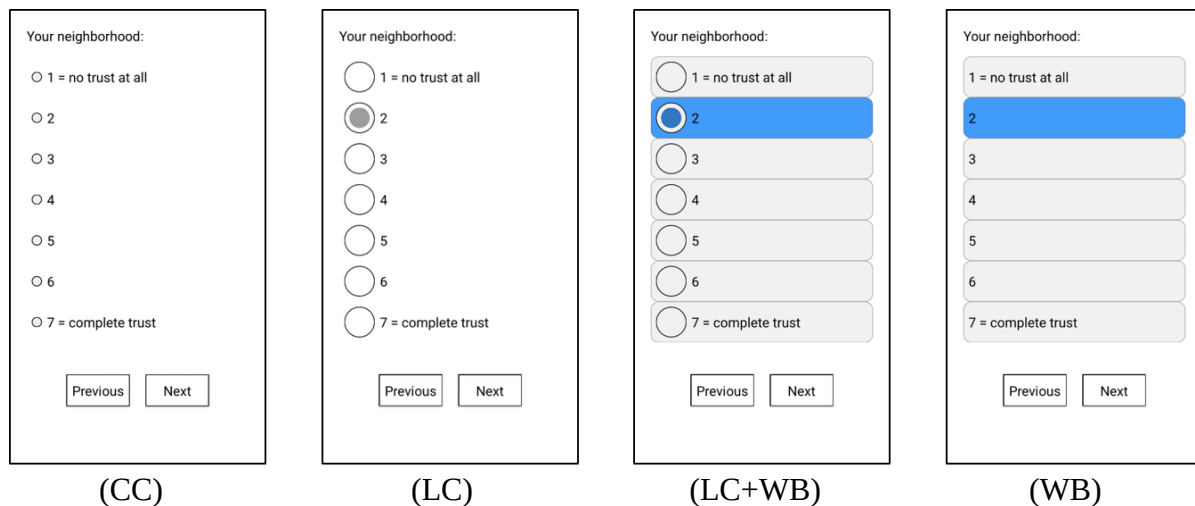


Figure 1. Four designs of response option: conventional controls (CC), larger controls (LC), larger controls enclosed in wide buttons (LC+WB), and wide buttons without any controls (WB).

4. Performance Measures

- Question-level completion times:* time from page load to selection of "next" button
- Misses (errors):* number of times a participant tapped the screen without selecting a response option
- Changed answers:* number of times a participant tapped the screen after their initial selection to change their answer

- d. *Number of categories selected*: number of categories selected for each choose-all-that-apply question
- e. *Satisfaction*: Participants' ratings of ease of completion and ease of answer selection
- f. *Preference*: Participants' overall preference among all designs that were tested

5. Experimental Factor

Response Option

- a. *Conventional controls (CC)*: small icons (2mm in diameter)
- b. *Larger controls (LC)*: large icons (6 mm in diameter)
- c. *Larger controls enclosed in wide buttons (LC+WB) "Hybrid buttons"*: wide buttons (that change color upon selection) containing large icons
- d. *Wide buttons without any controls (WB) "Plain buttons"*: wide buttons (that change color upon selection) that do not contain icons

6. Experimental Design

This was a between-subjects design. Participants were sequentially assigned (first participant to condition 1, second participant to condition 2, and so forth) to one of four conditions, CC, LC, LC+WB, and WB.

7. Instrument

iPhone 5s.

8. Experimental Procedure

Participants first completed a paper questionnaire containing demographic questions. Then a test administrator handed an iPhone 5s with a preloaded survey app to the participant and instructed him/her to complete the survey. After completing the survey on the iPhone, the participant completed a paper post-survey questionnaire with evaluation questions.

9. Data Analysis

For measures captured by the app there were 1,222 observations in total (47 participants x 26 pages). To compare the effects of the four response option designs on survey performance, we fit a linear mixed model (LMM) using the experimental conditions and a random effect of respondents to account for the hierarchical data structure (pages nested within respondents). Pairwise comparison results that were not statistically significant were not reported here.

Question-level timings were truncated at the 95th percentile to remove extreme values resulting in N=1,161. A sensitivity analysis of the log-transformed timings did not lead to changes in our conclusions.

Misses were aggregated across screens for each individual participant to produce a summary measure of the percentage of pages in which they made an errant tap.

Answer changes were aggregated across screens for each individual participant to produce a summary measure of the percentage of pages in which they changed their answer.

Results

Question-level completion time. There were no statistically significant differences in completion time using the other three formats (*LC*, *WB*, *LC+WB*) compared to the conventional controls ($p > .05$) (see Figure 2A).

Misses. As shown in Figure 2B, participants made significantly fewer misses using the optimized formats than the conventional controls (*LC* vs. *CC*, $p < .01$; *LC+WB* vs. *CC*, $p < .01$; and *WB* vs. *CC*, $p = .03$). These differences were relatively large in magnitude: participants made an errant tap on approximately 25% of pages when using the smaller controls compared to 10% of pages when using the wide buttons, 6% of pages when using the larger controls embedded in wide buttons, and 3% of pages when using the standalone larger controls. There are no statistically significant differences with respect to misses among the three optimized conditions.

Answer changes. Participants did not change their answers at a high rate in any of the conditions. And there was no statistically significant difference in answer changes among the four control formats (see Figure 2C)

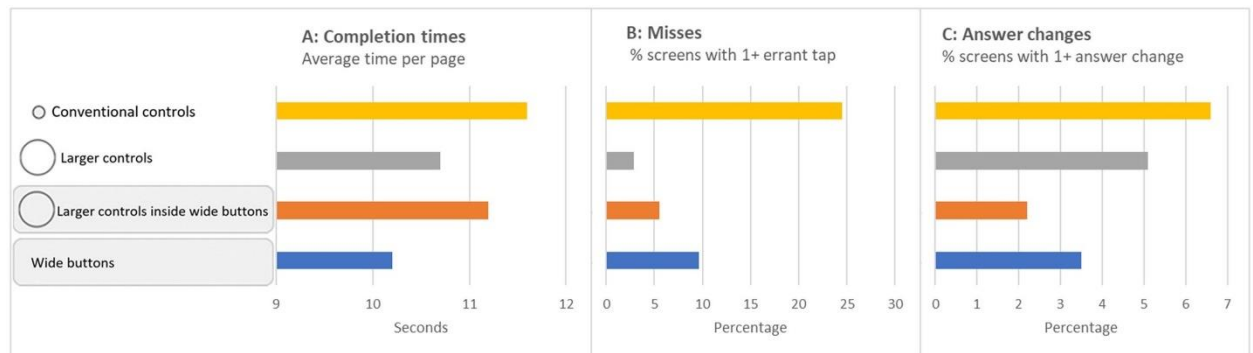


Figure 2. Performance measures across the four experimental conditions: (A) completion times; (B) misses; and (C) answer changes.

Number of categories selected. The use of wide buttons without any check boxes for choose-all questions raises the possibility that respondents would not realize that they could select more than one option. This did not appear to be the case. We found that participants selected approximately about the same number of categories across the four designs (*CC*: 5.1; *LC*: 5.3; *LC+WB*: 5.2; *WB*: 5.6) ($F = .24$, $p = .87$).

Respondent ratings and preferences. Since the unit of analysis for the post-survey evaluations is an individual participant ($n = 61$) rather than an individual survey page, the statistical power to detect differences is diminished compared to the earlier analyses. Still, the pattern of results suggests a preference for the larger controls, either on their own or enclosed in wide buttons. More participants who used one of these formats reported that it was “very easy” to select their answers (87% for *LC* and 80% for *LC+WB*, respectively) than those who used the plain wide buttons (63%) or conventional controls (60%) ($\chi^2(3) = 3.9$, $p = .28$). Similarly, after participants were shown all of the formats, most of them

reported a preference for the larger controls (54%) or larger controls embedded in wide buttons (36%) over the plain buttons (5%) or conventional controls (5%). The two most preferred formats did not differ from one another ($p = .13$) but did differ reliably from each of the other two designs ($p < .01$ for each pairwise comparison)

Recommendation

The present study compared usability among four response option designs: (1) small radio button or checkboxes (2 mm in diameter/width) (CC design), (2) large radio buttons or checkboxes (6 mm in diameter/width) (LC design), (3) large radio buttons or checkboxes embedded in wide buttons covering the response text (LC+WB design), and (4) plain wide buttons covering the response text (WB design). The findings show that participants made the least errors when tapping the LC design than the other designs; and more participants preferred the LC design than the other designs.

We recommend using radio button or check box with a 6-mm diameter/width (LC design) in mobile surveys.